

Project Design Phase-I

Solution Architecture

Date	28 February 2026
Team ID	LTVIP2026TMIDS88469
Project Name	Heart Disease Analysis
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture of the Heart Disease Analysis project is designed to deliver meaningful insights using Tableau dashboards.

1. Dataset Input :

The dataset contains patient information such as:

- Age
- Sex
- Chest Pain Type (cp)
- Resting Blood Pressure (trestbps)
- Cholesterol (chol)
- Fasting Blood Sugar (fbs)
- Resting ECG (restecg)
- Maximum Heart Rate (thalach)
- Exercise-induced Angina (exang)
- ST Depression (oldpeak)
- Heart Disease Target (0/1)

2. Data Cleaning & Preprocessing :

- Null values handled
- Data types corrected
- Duplicate records removed
- Feature columns standardized
- Exported cleaned dataset for Tableau

3. Data Storage :

- Dataset stored in CSV
- Imported into **MySQL database** for querying and analysis

4. Dashboard Development (Tableau) :

Multiple worksheets created such as:

- Age group vs heart disease
- Gender distribution
- Cholesterol vs heart disease
- Chest pain type analysis
- Blood pressure analysis
- Max heart rate patterns

5. Publishing & Sharing :

Dashboards and Story published in Tableau Public and embedded into Flask web app.

6. End Users :

- Students / Analysts
- Healthcare Researchers
- Doctors (for understanding patterns)